

NWIS

WELL COMPLETION REPORT

Doc ID
W004-WCR-01

Well: Alpha-04 | Well ID: W004 | Field: Field_Alpha

PRIMARY ISSUE CATEGORY: CEMENTING FAILURE

1. GENERAL WELL INFORMATION

Well Name	Alpha-04	Well ID	W004
Field	Field_Alpha	Block	Block_2
Spud Date	05-11-2024	Rig Release Date	02-12-2024
Total Depth (TD)	3120 m	Rig Name	Rig Sagar-4
Well Type	Vertical Development	Trajectory	Vertical
Max Inclination	1.5 deg	Directional Interval	N/A (near-vertical)

2. FORMATION & PORE PRESSURE INFORMATION

The intermediate casing shoe was set at 2600 m within Formation_X (lower), a zone with alternating high-permeability sand streaks that had already caused partial losses during drilling of the 12-1/4 in hole section. This loss history was the direct precursor to the cementing difficulties described below.

3. LITHOLOGY SUMMARY

Interval (m)	Formation	Lithology	Notes
2300-2550	Formation_W	Shale, grey, firm	Normal drilling trend
2550-2600	Formation_X (upper)	Sand, friable, thin streaks	Loss history during drilling; shoe set here
2600-2850	Formation_X (mid)	Shale/sand interbedded	Normal drilling trend
2850-3120	Formation_X (lower)	Sand, friable, high permeability	Loss-prone; monitor returns

4. HISTORICAL DRILLING / OPERATIONS EVENTS

Event 1: Cementing Failure (Losses During Job / Poor Bond) at approximately 2600 m (shoe)

Observation: During the 9-5/8 in intermediate casing cement job, full returns were lost approximately 40 minutes into pumping the tail slurry; an estimated 55 bbl of cement was lost to the loss-prone sand streak immediately above the shoe. Post-job cement bond log (CBL) and variable density log (VDL) showed poor bond quality (amplitude attenuation <30%) across 2540-2600 m with an annular gas channel interpreted at 2565-2580 m.

Engineering Action: Job was continued to completion with reduced pump rate once losses were detected, followed by a top-up job. Because the CBL/VDL confirmed an inadequate bond, a remedial squeeze cementing operation was performed at 2570 m using 25 bbl of Class G neat slurry (15.8 ppg) with a perforate-and-squeeze technique through the affected interval.

Outcome: Follow-up CBL/VDL log after the squeeze job showed improved bond quality (amplitude attenuation >65%) across the treated interval, confirmed by a successful casing pressure test to 3500 psi. Total NPT for the original job losses plus remedial squeeze was 34 hours.

5. CASING AND COMPLETION SUMMARY

Casing	Size (in)	Set Depth (m)	Cement Top (m)
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Conductor	20	56	Surface
Surface	13-3/8	630	Surface
Intermediate	9-5/8	2600	1850 (design) / 2540 (actual, pre-squeeze)
Production	7	3120	2250

6. NON-PRODUCTIVE TIME (NPT) SUMMARY

Issue	Depth (m)	NPT (hrs)	Category
Cement losses during primary job	2600	10.0	Cementing / Lost Circulation
Remedial squeeze cementing	2570	24.0	Cementing / Remedial

7. ROOT CAUSE ANALYSIS

Root cause analysis concluded that the primary cement job was pumped without adequate lost-circulation pre-treatment despite a known loss history in the 12-1/4 in section immediately above the shoe. The tail slurry design (15.8 ppg, standard Class G, no LCM additive) was not adjusted for the known loss-prone interval, and the resulting fluid loss into the formation created an under-filled annulus and channeling at 2565-2580 m.

8. LESSONS LEARNED

- Any casing shoe set within or immediately below the Formation_X sand streaks should use a lead slurry with LCM additive (e.g., 1/4-1/2 lb/sk cellophane flake or equivalent).
- Stage cementing or a lighter lead slurry (13.5-14.0 ppg) should be considered for shoes in this interval to reduce hydrostatic overbalance against the loss zone.
- CBL/VDL logging should be run as standard practice (not contingent) for any shoe set in a known loss interval, given the elevated channeling risk demonstrated here.
- Pre-job loss circulation material should be spotted across the loss interval before cementing begins, not only during drilling.

9. RECOMMENDATION FOR OFFSET WELLS

For offset wells setting casing shoes within the Formation_X loss-prone sand (2550-2950 m), specify an LCM-additive lead slurry, reduce lead slurry density to 13.5-14.0 ppg, spot an LCM pill immediately ahead of the cement job, and make CBL/VDL logging mandatory rather than optional.

This Well Completion Report is a synthetically generated historical record produced for the NWIS prototype (W004). Not associated with an actual drilling operation.